



University of Tehran
University College of Engineering
School of Electrical and Computer Engineering



Machine Learning

HW 1

Parsa Kafshduzibukani

SID: 810102501

Aban 1404

Table of Contents

LIST OF FIGURES	4
SUMMARY.....	5
QUESTION 1 - BAYES RULE VS. RANDOM DECISIONS	6
1.1. OVERALL RISK	6
1.2. R_{BAYES} V.S. R_{RAND}	6
1.3. WHEN ARE THEY EQUAL?	7
QUESTION 2 - BAYES BOUNDARIES FOR GAUSSIAN CLASSES ...	7
2.1. THRESHOLD CALCULATION	7
2.2. (A).....	8
2.2. (B).....	8
2.2. (C)	9
2.2. (D)	9
QUESTION 3 - NAIVE BAYES PARAMETER ESTIMATION (MLE)	10
3.1. JOINT LOG-LIKELIHOOD	10
3.2. EMPIRICAL FREQUENCIES	10
QUESTION 4 - MODELING GOALS WITH MLE	11
4.1. MAXIMUM-LIKELIHOOD ESTIMATION	11
4.2. PROBABILITY OF SCORING 6 GOALS	12
4.3. UPDATE λ , ESTIMATE AGAIN.....	12
QUESTION 5 - MLE DERIVATION OF THE LEAST SQUARES LOSS.....	13
5.1. CALCULATE LIKELIHOOD $p(y_i x_i; \theta, \sigma^2)$	13
5.2. MLE'S CONNECTION WITH MSE	13
QUESTION 6 - EFFECT OF PRIORS ON THE DECISION BOUNDARY	13
QUESTION 7 - NAIVE BAYES WARM-UP	14
7.1. DRUG-TEST PROBABILITY.....	14
7.2. NAIVE BAYES CLASSIFIER IMPLEMENTATION	14
QUESTION 8: CLASSIFYING SPAM WITH NAIVE BAYES	15
8.2. COMPUTE CLASS PRIORS.....	15
8.3. CONDITIONAL WORD PROBABILITIES.....	15
8.4. CLASSIFYING TEST DATA.....	16

8.5. EFFECT OF THE M PARAMETER	16
8.6. EMAILS TO BEAT THE CLASSIFIERS	17
QUESTION 9 - CLOUD VS. CLEAR SKY	17
9.1. DEFINING A CLASSIFICATION CRITERION	17
9.2. EVALUATION METRICS	17
9.3. IMPROVED CLASSIFIER USING BRIGHTNESS.....	18
9.4. MISCLASSIFIED IMAGES AND DISCUSSION	19
9.5. COMPARISON OF THE CLASSIFIERS.....	20
REFERENCES.....	21

List of figures

FIGURE 1 - Bayes Decision Boundary	9
FIGURE 2 - Effect of the m Parameter.....	17
FIGURE 3 - Feature Distribution.....	19
FIGURE 4 - Examples of Misclassified Images	19

Summary

The purpose of this set of exercises was to explore fundamental principles of Bayesian decision theory, maximum likelihood estimation (MLE), and simple probabilistic classification in practical settings.

Throughout the homework, several key machine learning concepts were applied: deriving Bayes decision rules, computing risks and posterior probabilities, estimating parameters for Bernoulli and Poisson models, and implementing interpretable classifiers such as the Naive Bayes algorithm and a color-based image classifier for cloudy versus clear skies.

Question 1 - Bayes Rule vs. Random Decisions

1.1. Overall Risk

Suppose the task is to classify the input signal x into one of K classes $\omega \in \{1, 2, \dots, K\}$ such that the action $\alpha(x) = i$ means classifying x into class i .

The Bayesian decision rule is

$$\alpha_{\text{Bayes}}(x) = \omega^* = \arg \max_i P(\omega_i|x)$$

Suppose we replace it by a randomized decision rule which classifies x to class i following the posterior probability $P(\omega_i|x)$, i.e. $\alpha_{\text{rand}}(x) = \omega \sim P(\omega|x)$.

The conditional risk for deciding class i is

$$R(a_i|x) = \sum_{j=1}^K l(a_i|\omega_j)P(\omega_j|x)$$

Using the zero-one loss, where $l(a_i|\omega_j) = 1$ for $i \neq j$ and 0 otherwise, we obtain

$$R(a_i|x) = 1 - P(\omega_i|x)$$

In the randomized rule, class i is chosen with probability $P(\omega_i|x)$, thus

$$\begin{aligned} R_{\text{rand}}(x) &= \sum_{i=1}^K P(\text{choose } i|x) R(a_i|x) \\ &= \sum_{i=1}^K P(\omega_i|x)[1 - P(\omega_i|x)] \\ &= 1 - \sum_{i=1}^K P(\omega_i|x)^2 \end{aligned}$$

The overall expected risk is therefore

$$R_{\text{rand}} = \int \left[1 - \sum_{i=1}^K P(\omega_i|x)^2 \right] p(x) dx$$

1.2. $R_{\text{Bayes}}(x)$ V.S. $R_{\text{rand}}(x)$

For the Bayes decision rule, the conditional risk at a given x is

$$R_{\text{Bayes}}(x) = 1 - \max_i P(\omega_i|x)$$

and for the randomized rule it is

$$R_{\text{rand}}(x) = 1 - \sum_{i=1}^K P(\omega_i|x)^2$$

Since all $P(\omega_i|x)$ are between 0 and 1 and sum to 1, the **largest** of them is always at least as big as the **sum of their squares**. Thus

$$\max_i P(\omega_i|x) \geq \sum_{i=1}^K P(\omega_i|x)^2$$

Subtract both sides from 1 we have

$$R_{\text{rand}}(x) \geq R_{\text{Bayes}}(x)$$

Hence, the randomized decision rule never yields a smaller risk than the deterministic Bayes rule.

1.3. When are they equal?

The Bayes decision rule chooses the class with the largest posterior probability, while the randomized rule samples from the posterior distribution itself.

They produce identical results only when all posterior probabilities are equal, i.e.,

$$P(\omega_1|x) = P(\omega_2|x) = \dots = P(\omega_K|x)$$

In this case, no class is preferred over another, so the Bayes rule may choose any class and the randomized rule selects uniformly.

Question 2 - Bayes Boundaries for Gaussian Classes

2.1. Threshold Calculation

From the Bayes decision rule,

$$\frac{p(x|\omega_1)}{p(x|\omega_2)} \geq_{\omega_1} \frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)}$$

Substituting the Gaussian densities,

$$\frac{\exp[-x^2/(2\sigma^2)]}{\exp[-(x-1)^2/(2\sigma^2)]} = \frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)}$$

Simplifying the exponent,

$$\begin{aligned} \exp\left(\frac{(x-1)^2 - x^2}{2\sigma^2}\right) &= \frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)} \\ \exp\left(\frac{-2x+1}{2\sigma^2}\right) &= \frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)} \end{aligned}$$

Taking the logarithm and solving for x,

$$x = \frac{1}{2} - \sigma^2 \ln\left(\frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)}\right)$$

Hence, the optimal threshold is

$$\tau = \frac{1}{2} - \sigma^2 \ln\left(\frac{\lambda_{12}P(\omega_2)}{\lambda_{21}P(\omega_1)}\right)$$

2.2. (a)

For Gaussian classes with equal covariance matrix Σ , the discriminant functions are

$$g_i(x) = x^T \Sigma^{-1} \mu_i - \frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \ln P(\omega_i)$$

Given $\Sigma = 3I$, we have $\Sigma^{-1} = \frac{1}{3}I$, hence

$$g_i(x) = \frac{1}{3} x^T \mu_i - \frac{1}{6} \mu_i^T \mu_i + \ln P(\omega_i)$$

Substituting the class parameters:

$$\begin{aligned} g_1(x) &= \frac{1}{3}(4x_1 + 11x_2) - \frac{1}{6}(4^2 + 11^2) + \ln(0.6) \\ g_2(x) &= \frac{1}{3}(10x_1 + 3x_2) - \frac{1}{6}(10^2 + 3^2) + \ln(0.4) \end{aligned}$$

The Bayes decision rule is: decide ω_1 if $g_1(x) > g_2(x)$, otherwise decide ω_2 .

2.2. (b)

The decision boundary satisfies

$$g_1(x) = g_2(x)$$

Using

$$g_i(x) = \frac{1}{3} x^T \mu_i - \frac{1}{6} \mu_i^T \mu_i + \ln P(\omega_i),$$

we have

$$\frac{1}{3} x^T (\mu_1 - \mu_2) = \frac{1}{6} (\mu_1^T \mu_1 - \mu_2^T \mu_2) + \ln \frac{P(\omega_2)}{P(\omega_1)}.$$

With $\mu_1 = [4, 11]^T$, $\mu_2 = [10, 3]^T$, and $P(\omega_1) = 0.6$, $P(\omega_2) = 0.4$:

$$\begin{aligned} \mu_1 - \mu_2 &= [-6, 8]^T, \\ \mu_1^T \mu_1 - \mu_2^T \mu_2 &= 28. \end{aligned}$$

Hence,

$$\frac{1}{3}(-6x_1 + 8x_2) = \frac{1}{6}(28) + \ln \frac{0.4}{0.6}$$

Simplifying,

$$-6x_1 + 8x_2 = 14 + 3 \ln \frac{2}{3}.$$

Therefore, the linear decision boundary is

$$6x_1 - 8x_2 + 14 + 3 \ln \frac{2}{3} = 0.$$

2.2. (c)

Effect of Priors:

The priors appear in the logarithmic term of the discriminant functions,

$$g_i(x) = x^T \Sigma^{-1} \mu_i - \frac{1}{2} \mu_i^T \Sigma^{-1} \mu_i + \ln(P(\omega_i))$$

Changing the priors adds a constant offset to each discriminant function.

Increasing $P(\omega_1)$ shifts the decision boundary toward class ω_2 , expanding the region for ω_1 , and vice versa.

Effect of Covariances:

If the covariance matrices are equal, the decision boundary is linear.

When the covariances differ, the discriminant functions become quadratic:

$$g_i(x) = -\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) - \frac{1}{2} \ln(|\Sigma_i|) + \ln(P(\omega_i))$$

Thus, the boundary $g_1(x) = g_2(x)$ becomes a *quadratic curve* rather than a straight line.

2.2. (d)

The discriminant functions are

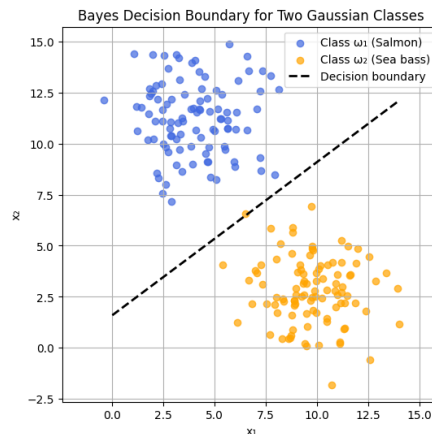
$$g_1(x) = \frac{1}{3} (4x_1 + 11x_2) - \frac{1}{6} (4^2 + 11^2) + \ln(0.6),$$

$$g_2(x) = \frac{1}{3} (10x_1 + 3x_2) - \frac{1}{6} (10^2 + 3^2) + \ln(0.4).$$

The decision boundary obtained from $g_1(x) = g_2(x)$ is

$$6x_1 - 8x_2 + 14 + 3 \ln(2/3) = 0$$

When plotted, the samples from class ω_1 form a cluster around (4, 11) and those from class ω_2 around (10, 3). The straight line defined by the above equation separates the two clusters and is slightly shifted toward class ω_2 , the less probable class.



Question 3 - Naive Bayes Parameter Estimation (MLE)

3.1. Joint Log-Likelihood

For m training examples $(x^{(i)}, y^{(i)})$, the Naive Bayes model assumes conditional independence:

$$P(x^{(i)}, y^{(i)}; \phi) = P(y^{(i)}; \phi_y) \prod_{j=1}^n P(x_j^{(i)} | y^{(i)}; \phi_{j|y})$$

Since each feature is binary ($x_j \in \{0, 1\}$), it follows a Bernoulli distribution:

$$P(x_j^{(i)} | y^{(i)}; \phi_{j|y^{(i)}}) = (\phi_{j|y^{(i)}})^{x_j^{(i)}} (1 - \phi_{j|y^{(i)}})^{1-x_j^{(i)}}$$

Substitute this back into the joint expression to get:

$$P(x^{(i)}, y^{(i)}; \phi) = P(y^{(i)}; \phi_y) \prod_{j=1}^n \left[\phi_{j|y^{(i)}}^{x_j^{(i)}} (1 - \phi_{j|y^{(i)}})^{1-x_j^{(i)}} \right]$$

For m training samples, the overall likelihood is the product of all individual sample probabilities:

$$L(\phi) = \prod_{i=1}^m P(x^{(i)}, y^{(i)}; \phi)$$

Taking the natural logarithm (to simplify multiplication into addition)

$$\ell(\phi) = \sum_{i=1}^m \ln P(x^{(i)}, y^{(i)}; \phi)$$

$$\ell(\phi) = \sum_{i=1}^m \left[\ln P(y^{(i)}; \phi_y) + \sum_{j=1}^n \left(x_j^{(i)} \ln \phi_{j|y^{(i)}} + (1 - x_j^{(i)}) \ln(1 - \phi_{j|y^{(i)}}) \right) \right]$$

Expressing the prior term explicitly

$$\ln P(y^{(i)}; \phi_y) = y^{(i)} \ln \phi_y + (1 - y^{(i)}) \ln(1 - \phi_y)$$

Substitute this into the previous expression to get the final **joint log-likelihood**:

$$\ell(\phi) = \sum_{i=1}^m \left[y^{(i)} \ln \phi_y + (1 - y^{(i)}) \ln(1 - \phi_y) + \sum_{j=1}^n \left(x_j^{(i)} \ln \phi_{j|y^{(i)}} + (1 - x_j^{(i)}) \ln(1 - \phi_{j|y^{(i)}}) \right) \right]$$

3.2. empirical frequencies

By maximizing $\ell(\phi)$ with respect to each parameter:

For ϕ_y :

$$\frac{\partial \ell}{\partial \phi_y} = \frac{\sum_i y^{(i)}}{\phi_y} - \frac{\sum_i (1 - y^{(i)})}{1 - \phi_y} = 0$$

which gives

$$\phi_{-y} = \frac{\sum_i y^{(i)}}{m}$$

For the parameter $\phi_{j|y=1}$, we keep only the terms in the log-likelihood that depend on it:

$$\sum_{i:y^{(i)}=1} x_j^{\{i\}} \ln(\phi_{j|y=1}) + (1 - x_j^{\{i\}}) \ln(1 - \phi_{j|y=1})$$

Taking the derivative with respect to $\phi_{j|y=1}$ and setting it equal to zero gives:

$$\phi_{j|y=1} = \frac{\sum_{i:y^{(i)}=1} x_j^{\{i\}}}{\sum_i y^{(i)}}$$

which is the empirical frequency of feature j being 1 among all samples where $y = 1$.

Following the same steps for samples with $y = 0$, gives:

$$\phi_{j|y=0} = \frac{\sum_{i:y^{(i)}=0} x_j^{\{i\}}}{\sum_i (1 - y^{(i)})}$$

which represents the empirical probability of feature j being 1 among all samples where $y = 0$.

Question 4 - Modeling Goals with MLE

4.1. Maximum-likelihood Estimation

Let $x_1, x_2, \dots, x_n$ represent the number of goals scored in n games, each modeled as an independent Poisson random variable with parameter λ .

The probability mass function is

$$P(x_i|\lambda) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

The likelihood function is

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} \\ &= \frac{(e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i})}{(\prod_{i=1}^n x_i!)} \end{aligned}$$

Taking the logarithm,

$$l(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i \right) \ln(\lambda) - \sum_{i=1}^n \ln(x_i!)$$

Differentiating with respect to λ and setting the derivative equal to zero,

$$\frac{dl}{d\lambda} = -n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0$$

Thus, the maximum-likelihood estimate is

$$\widehat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i$$

Therefore, the MLE of λ equals the sample mean of the observed goal counts.

4.2. Probability of scoring 6 goals

From Question 4.1, the MLE of λ is

$$\widehat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i$$

Using the five observed goal counts [2, 4, 6, 0, 1],

$$\hat{\lambda} = \frac{2 + 4 + 6 + 0 + 1}{5} = \frac{13}{5} = 2.6$$

For a Poisson random variable, the probability of observing $x=6$ goals is

$$P(X = 6|\lambda) = \frac{e^{-\lambda}\lambda^6}{6!}$$

$$P(X = 6|2.6) = \frac{e^{-2.6} \times 2.6^6}{6!} \approx 0.036$$

Hence, the probability that the **Reign score six goals in their next game is approximately 3.6%.**

4.3. Update λ , estimate again

Including the sixth game, the observed goal counts are [2, 4, 6, 0, 1, 8].

The maximum-likelihood estimate is

$$\widehat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{2 + 4 + 6 + 0 + 1 + 8}{6} = \frac{21}{6} = 3.5.$$

For a Poisson random variable, the probability of observing $x=6$ goals is

$$P(X = 6|3.5) = \frac{e^{-3.5} \times 3.5^6}{6!} \approx 0.073.$$

Therefore, **the probability that the Reign score six goals in their seventh game is approximately 7.3%.**

Question 5 - MLE Derivation of the Least Squares Loss

5.1. Calculate likelihood $p(y^{(i)}|x^{(i)}; \theta, \sigma^2)$

Given the model

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)}, \text{ where } \epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2),$$

the conditional distribution of $y^{(i)}$ given $x^{(i)}$ is Gaussian (since $\epsilon^{(i)}$ is *gaussian* then adding a constant ($\theta^T x^{(i)}$) shifts the mean by that constant):

$$y^{(i)} | x^{(i)}, \theta, \sigma^2 \sim N(\theta^T x^{(i)}, \sigma^2)$$

Therefore, the likelihood for one observation is

$$p(y^{(i)}|x^{(i)}; \theta, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

5.2. MLE's Connection with MSE

Assuming independence of all n samples, the joint likelihood is

$$L(\theta) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2\right)$$

Taking the logarithm,

$$l(\theta) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

Since the first term does not depend on θ and σ^2 is constant, maximizing $l(\theta)$ is equivalent to minimizing

$$\sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

which is the MSE formulation.

Hence, maximizing the log-likelihood corresponds exactly to minimizing the Mean Squared Error cost function used in linear regression.

Question 6 - Effect of Priors on the Decision Boundary

For a one-dimensional two-class problem with

$$p(x|\omega_1) = N(\mu_1, \sigma^2),$$

$$p(x|\omega_2) = N(\mu_2, \sigma^2),$$

the Bayes decision rule is

$$p(x|\omega_1)P(\omega_1) \geq_{\omega_2^1} p(x|\omega_2)P(\omega_2)$$

Taking logarithms(since the log is a monotonic function, the inequality remains the same),

$$\ln P(\omega_1) - \frac{(x - \mu_1)^2}{2\sigma^2} \geq_{\omega_2^1} \ln P(\omega_2) - \frac{(x - \mu_2)^2}{2\sigma^2}$$

Simplifying,

$$\frac{2x(\mu_1 - \mu_2) + (\mu_2^2 - \mu_1^2)}{2\sigma^2} = \ln\left(\frac{P(\omega_2)}{P(\omega_1)}\right)$$

Solving for the boundary point x^* ,

$$x^* = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_1 - \mu_2} \ln\left(\frac{P(\omega_2)}{P(\omega_1)}\right)$$

If $P(\omega_1) = P(\omega_2)$, the boundary lies midway between the means: $x^* = (\mu_1 + \mu_2)/2$

If $P(\omega_1) > P(\omega_2)$, the term $\ln(P(\omega_2)/P(\omega_1))$ is negative, and x^* shifts toward ω_2 the classifier **favors ω_1** .

If $P(\omega_1) < P(\omega_2)$, the term is positive, and x^* shifts toward ω_1 favoring ω_2 .

Question 7 - Naive Bayes Warm-Up

7.1. Drug-Test Probability

Using Bayes' rule,

$$P(D = 1|T = 1) = \frac{P(T = 1|D = 1) P(D = 1)}{P(T = 1|D = 1) P(D = 1) + P(T = 1|D = 0) P(D = 0)}$$

Since $P(D = 0) = 1 - P(D = 1) = 0.999$ and $P(T = 1|D = 0) = 1 - P(T = 0|D = 0) = 0.01$, we have

$$P(D = 1|T = 1) = \frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.01 \times 0.999} = \frac{0.00099}{0.01098} \approx 0.09$$

Although the test has 99% accuracy, the probability that a person who tests positive actually uses the drug is only about 9%, because drug use is very rare in the population.

7.2. Naive Bayes Classifier Implementation

The prior probabilities are estimated as

$$P(y = 1) = \frac{\text{number of samples with } y = 1}{n}, \quad P(y = 0) = 1 - P(y = 1)$$

For each feature x_j and class $y=c$,

$$P(x_j = 1|y = c) = \frac{\text{number of samples where } x_j = 1 \text{ and } y = c}{\text{number of samples where } y = c}$$

For a test example $x = [1,1]$, the posterior probabilities are proportional to

$$P(y = c|x) \propto P(y = c) \prod_{j=1}^2 P(x_j|y = c)$$

The predicted label is the class with the higher posterior probability.

Question 8: Classifying Spam with Naive Bayes

8.2. Compute Class Priors

To estimate the class priors, we count how many training examples are labeled as spam and ham.

$$P(\text{spam}) = \frac{\text{number of spam emails}}{\text{total emails}}$$

$$P(\text{ham}) = 1 - P(\text{spam})$$

These prior probabilities indicate how likely an email is to be spam or ham before observing any of its words.

8.3. Conditional Word Probabilities

We construct a vocabulary V from all unique words in the training corpus and count their frequencies.

in spam and ham messages:

$$n_i^{(\text{spam})} = \text{number of occurrences of word } w_i \text{ in spam emails}$$

$$n_i^{(\text{ham})} = \text{number of occurrences of word } w_i \text{ in ham emails}$$

Let N_{spam} and N_{ham} be the total number of word tokens in spam and ham messages.

Using the m-estimate with $m = |V|$ and $p = 1/|V|$, the conditional probabilities are given by

$$P(w_i|\text{spam}) = \frac{n_i^{(\text{spam})} + mp}{N_{\text{spam}} + m},$$

$$P(w_i|\text{ham}) = \frac{n_i^{(\text{ham})} + mp}{N_{\text{ham}} + m}$$

After computing these probabilities, we rank the words according to their likelihood within each class (to identify the five most probable words for spam and for ham emails).

8.4. Classifying Test Data

Using the previously estimated probabilities, each test email is classified by comparing the log-posterior probabilities:

$$\log P(\text{spam}|x) \propto \log P(\text{spam}) + \sum_i n_i \log P(w_i|\text{spam}),$$

$$\log P(\text{ham}|x) \propto \log P(\text{ham}) + \sum_i n_i \log P(w_i|\text{ham}),$$

where n_i is the number of occurrences of word w_i in the email.

The predicted class is the one with the higher log-probability.

Model performance is evaluated using classification accuracy:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of test samples}}$$

8.5. Effect of the m Parameter

To study the influence of the smoothing parameter m , we vary it as

$$m = |V| \times [1, 10, 100, 1000, 10000]$$

For each value, we recompute the conditional probabilities

$$P(w_i|\text{spam}) = \frac{n_i^{(\text{spam})} + mp}{N_{\text{spam}} + m},$$

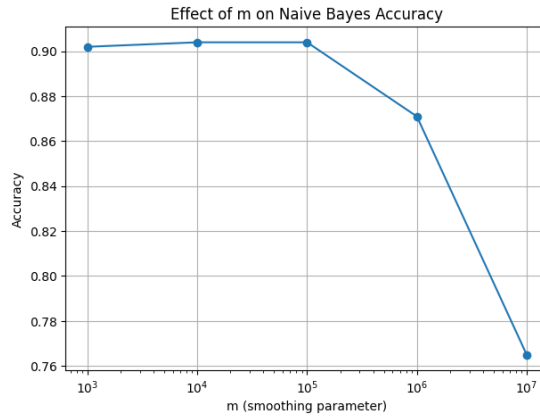
$$P(w_i|\text{ham}) = \frac{n_i^{(\text{ham})} + mp}{N_{\text{ham}} + m},$$

Where $p = 1/|V|$, and evaluate classification accuracy on the test set.

For very small m , the model relies almost entirely on observed word counts and may overfit to rare words.

For very large m , probabilities approach the uniform prior p , producing excessive smoothing and underfitting.

Accuracy typically peaks at an intermediate m , balancing bias and variance.



8.6. Emails to Beat the Classifiers

Adversaries can exploit Naive Bayes' assumptions (bag-of-words, exact token matches, and reliance on training frequencies) to reduce spam scores. Typical high-level strategies are: (1) **Obfuscation** - alter tokens so they no longer match high-spam vocabulary; (2) **Dilution** - add large amounts of neutral/ham-like text to overwhelm spam signals; (3) **Use of rare or out-of-vocabulary words** to rely on uniform smoothing probabilities.

Question 9 - Cloud vs. Clear Sky

9.1. Defining a Classification Criterion

As an initial baseline, we define a simple and interpretable criterion that relies only on the **mean saturation** of each image in the HSV color space.

Cloudy images tend to have dull, grayish colors and thus low saturation values, while clear-sky images are more colorful and saturated due to strong blue tones.

Decision rule:

If $\bar{S} < T_S$, classify as *cloudy*, otherwise classify as *clear*.

The threshold T_S is chosen empirically from the dataset distribution.

Although simple, this baseline provides an interpretable reference for later improvements.

9.2. Evaluation Metrics

Using the threshold $T_S = 0.42$ each image is classified as cloudy or clear according to

If $\bar{S} < T_S$, classify as cloudy, otherwise classify as clear.

Model performance is evaluated using:

$$\text{Precision} = \frac{TP}{TP + FP},$$

$$\text{Recall} = \frac{TP}{TP + FN},$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

A confusion matrix summarizes the counts of correctly and incorrectly classified samples.

Results:

Confusion Matrix:

$$\begin{bmatrix} 31 & 9 \\ 5 & 35 \end{bmatrix}$$

Precision (Cloudy class): 0.861

Recall (Cloudy class): 0.775

Overall Accuracy: 0.825

9.3. Improved Classifier Using Brightness

To improve upon the baseline, we extend the previous rule by incorporating both **saturation** and **brightness** from the HSV color space.

While saturation alone captures color vividness, cloudy scenes are not only less saturated but

also darker. Including brightness allows the classifier to separate overcast but bright conditions from truly clear skies.

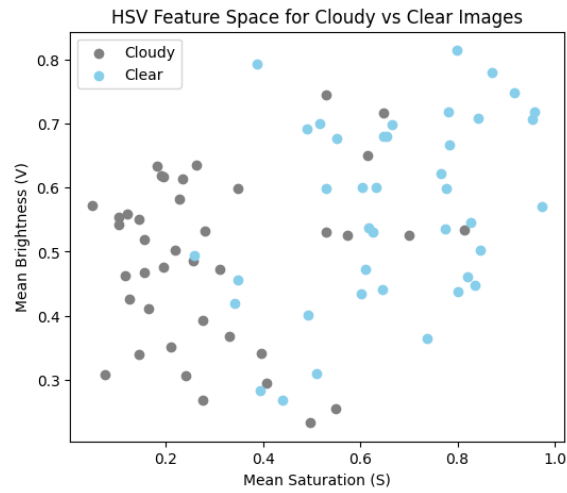
We apply the two-feature Decision rule:

If $\bar{S} < T_S$ and $\bar{V} < T_V$, classify as *cloudy*, otherwise classify as *clear*.

The thresholds T_S and T_V are determined empirically by observing the feature distribution

of the dataset.

This combined rule improves separability, capturing the intuitive observation that cloudy skies are both **darker** and **less saturated** than clear skies.



Results:

Confusion Matrix:

$\begin{bmatrix} 31 & 9 \end{bmatrix}$

$\begin{bmatrix} 4 & 36 \end{bmatrix}$

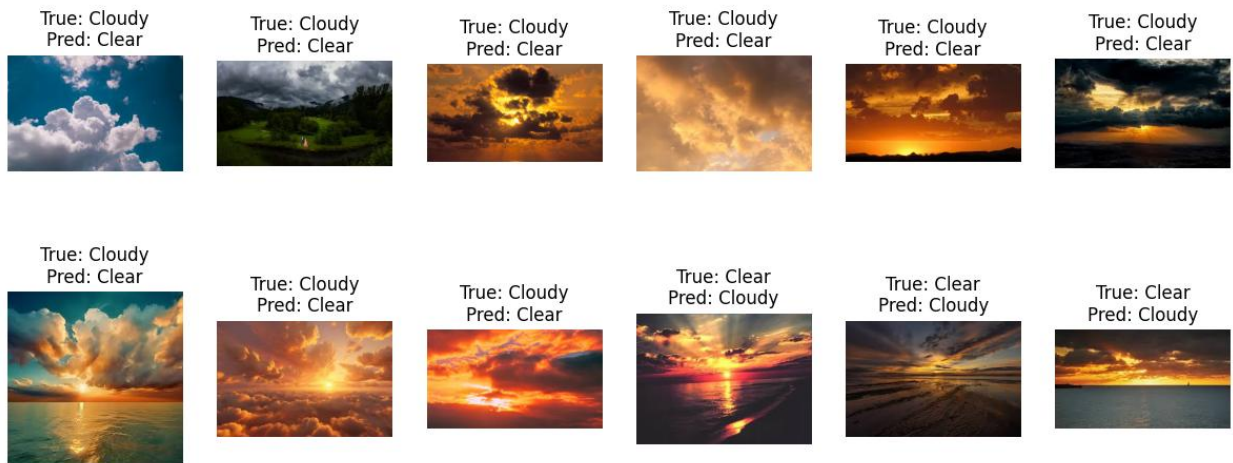
Precision (Cloudy class): 0.886

Recall (Cloudy class): 0.775

Overall Accuracy: 0.838

9.4. Misclassified Images and Discussion

Examples of Misclassified Images



To better understand the model behavior, the misclassified images were visualized.

The following trends were observed:

Partly cloudy scenes with bright sunlight or **blue sky patches** increase the average brightness and saturation, leading the model to predict *clear*.

Sunset and sunrise images, which are darker and less blue, have low saturation values and are incorrectly labeled as *cloudy*.

The current rule depends only on global color statistics (mean saturation and brightness), so intermediate conditions or unusual lighting can cause confusion.

9.5. Comparison of the Classifiers

Adding the brightness feature slightly improved both precision and overall accuracy, while recall for the cloudy class remained unchanged.

This suggests that incorporating brightness helped the model reduce false positives (clear images incorrectly labeled as cloudy), but did not yet address borderline cases where clouds and lighting conditions overlap.

Overall, the improved classifier achieves a more balanced decision boundary while remaining fully interpretable and simple to implement.

References